Subarray Divisibility

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1662 | Sorting and Searching | https://cses.fi/problemset/task/1662

Given an array of n integers, your task is to count the number of subarrays where the sum of values is divisible by n.

Input

The first input line has an integer n: the size of the array.

The next line has n integers $a_1,a_2,\ldots,a_n$: the contents of the array.

Output

Print one integer: the required number of subarrays.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $-10^9 \le a_i \le 10^9$

Example

Input
5 3 1 2 7 4

Output
1